Yuanbo Li

Brown University

PERSONAL INFORMATION

Yuanbo Li
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Department of Computer Science
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Providence, RI, USA 02912

Email: yuanbo_li@brown.edu
Homepage: https://liyb2002.github.io/
GitHub: https://github.com/Liyb2002
Google Scholar: https://scholar.google.com/citations?user=s_1CcsUAAAAJ&hl=en

RESEARCH INTERESTS

Computer Graphics, 3D Modeling, Computer-Aided-Design, Vision-Language Models, Neural-Symbolic Reasoning

EDUCATION

Ph.D. in Computer Science, Brown University — 2024–Present
Advisor: Daniel Ritchie

M.S. in Computer Science, Brown University — 2022–2024

B.A. in Mathematics, Columbia University — 2018–2022

PUBLICATIONS

Journal and Conference Papers

[J1] **Li, Y.**, Manfredi, G*., Kriel, H*., Hao, C., Xu, X., Bousseau, A., Ritchie, D.
CADrawer: Autoregressive CAD Generation from 3D Sketches. Computer Graphics Forum (Proc. Eurographics), 2026.

[J2] **Li, Y.**, Ma, T., Aljumayaat, Z., Ritchie, D.
PossibleImpossibles: Exploratory Procedural Design of Impossible Structures. Computer Graphics Forum (Proc. Eurographics), 2024.

Short Papers / Preprints

[S1] Kwon, H., **Li, Y.**, Ye, X., Muna-McQuay, P., Yin, L., Tompkin, J.
Active Appearance and Spatial Variation Can Improve Visibility in Area Labels for Augmented Reality. IEEE VIS, 2024.

[S2] **Li, Y.**, Shu, D., Chen, Y., Klenk, M., Ritchie, D.
PLLM: Pseudo Label Methods for LLM CAD Generation. Under submission.

[S3] Chang, A., Wang, K., **Li, Y.**, Savva, M., Chang, A. X., Ritchie, D.
Learning Object Placement Programs for Indoor Scene Synthesis with Iterative Self-Training. Arxiv.

EXPERIENCE

Autodesk Research — Summer 2026
Research Scientist Intern — CA, USA

Mentor(s): Vivian Liu, Joanne Leong, Mohi Reza, and John Thompson

Toyota Research Institute Summer 2025
Research Scientist Intern CA, USA
Mentor(s): Dule Shu, Yanying Chen, and Matt Klenk

ARPA Technology Summer 2022, Summer 2020
Software Engineer Intern NY, USA / Beijing, China

Tencent Summer 2021, Summer 2019
Product Manager Intern Beijing, China

PATENTS

[P1] **Pseudo-Label Learning Method for Training LLMs to Generate CAD Program Sequences**
Yuanbo Li, Dule Shu, Yanxia Zhang, Yanying Chen, Matt Klenk
U.S. Provisional Patent Application No. 63/863,263, Aug 2025.

STUDENT MENTORING

Xiaoxi Yang Research Assistant, Brown University

Chengye Hao M.S. in Computer Science, Brown University (expected 2026)

Bilal Tariq Undergraduate Student, Amherst College (expected 2027)

TEACHING

Teaching Assistant
Advanced Graphics, Brown University, Spring 2024

Teaching Assistant
Calculus III, Columbia University, Fall 2021

SELECTED TALKS

CAD Generation using LLMs, Toyota Research Institute, 2025
Sketch to CAD, Brown Visual Computing, 2024
Generating 3D Impossible Structures, Eurographics, 2024
Inverse Procedural Modeling, Brown Visual Computing, 2023

SERVICES

Reviewer, Siggraph (2026)
Reviewer, IEEE TVCG (2024)